Progress Report
Cost-Aware Probabilistic Temporal Shifting Framework for Carbon-Aware Cloud Data-Centres

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1. Project Overview
The increasing energy demand of large-scale cloud infrastructures has intensified the need for carbon-aware workload scheduling. This project proposes a **Cost-Aware Probabilistic Temporal Shifting Framework** that dynamically aligns delay-tolerant batch workloads with low-carbon electricity periods while minimizing economic penalties and SLA violations.
The system jointly optimizes environmental impact and financial feasibility using real-world workload traces, carbon intensity data, and cloud pricing histories.

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2. Objectives
1. To model the trade-off between carbon savings and reserved instance idling costs.
2. To develop a probabilistic scheduling algorithm that handles uncertain job durations.
3. To integrate real-world carbon and pricing traces into the optimization model.
4. To validate the framework against realistic cloud workloads.

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3. Data Collection and Integration
To ensure experimental credibility, three categories of real-world datasets have been incorporated.

3.1 Cloud Workload Traces (Job-Level Data)
a) Alibaba Group Cluster Trace
- Multi-day production cluster dataset
- Includes batch job submissions
- CPU and memory allocation details
- Machine-level utilization

This dataset is ideal for evaluating temporal shifting because it contains delay-tolerant batch workloads with varied execution lengths.
b) Microsoft Azure Public Workload Traces
- Azure 2019/2020 anonymized VM and function traces
- Highly dynamic and bursty workloads
- Realistic cloud elasticity behavior

These traces are used to validate robustness under highly variable job arrival patterns.

3.2 Carbon Intensity Data (Environmental Signal)
a) Electricity Maps
- Hourly grid carbon intensity (gCO_2/kWh)
- Region-specific emission signals
- Historical carbon traces

Used to align job execution windows with low-carbon periods.
b) WattTime
- Marginal emissions data
- Reflects actual carbon impact of incremental energy consumption

Provides additional validation for carbon-aware scheduling decisions.

3.3 Cloud Pricing Data (Economic Component)
a) Amazon Web Services Spot Price History
- Historical EC2 spot pricing
- Fluctuating demand-based pricing
- Used to model financial risk of waiting

b) Cloud Carbon Footprint

- Energy-to-carbon conversion coefficients
- Instance-level power consumption baselines
- Cost-to-energy mapping

Enables accurate financial and environmental cost modeling.

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4. System Model and Cost Formulation

The total operational objective function is defined as:

Ctotal=∑t=1T(αE(t)P(t)+βCI(t)E(t)+γV(t))C_{total} = \sum_{t=1}^{T} \left(\alpha E(t)P(t) + \beta CI(t)E(t) + \gamma V(t) \right)Ctotal=t=1∑T(αE(t)P(t)+βCI(t)E(t)+γV(t))

Where:

- E(t)E(t)E(t): Energy consumption
- P(t)P(t)P(t): Instance or electricity price
- CI(t)CI(t)CI(t): Grid carbon intensity
- V(t)V(t)V(t): SLA violation penalty
- α,β,γ\alpha, \beta, \gammaα,β,γ: Weighting coefficients

This formulation allows dynamic balancing between carbon reduction, economic cost, and performance guarantees.

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5. Work Completed

✓ Literature Review

- Surveyed carbon-aware scheduling strategies.
- Identified limitations in deterministic and fixed-threshold shifting.

✓ Dataset Acquisition

- Downloaded and preprocessed Alibaba and Azure workload traces.
- Integrated hourly carbon traces from Electricity Maps.
- Retrieved AWS spot price history.
- Calibrated energy coefficients using Cloud Carbon Footprint data.

✓ Algorithm Development

- Designed probabilistic decision-making mechanism.
- Implemented forecast-based shifting window selection.
- Integrated SLA-aware penalty modeling.
- Modeled idling cost of reserved instances.

✓ Baseline Implementation

Implemented comparison policies:

- No shifting (Immediate execution)
- Static threshold carbon shifting
- Greedy carbon-minimization strategy

✓ Simulation Framework

- Built discrete-time simulator.
- Integrated workload + carbon + pricing signals.
- Implemented performance evaluation metrics.

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6. Preliminary Results

Initial experiments demonstrate:

- Significant carbon emission reduction compared to no-shifting baseline.
- Reduced SLA violations compared to aggressive carbon-only strategies.
- Improved cost-efficiency compared to static threshold approaches.
- Balanced reserved instance utilization.

Early findings confirm that probabilistic cost-aware scheduling outperforms deterministic temporal shifting in realistic cloud scenarios.

7. Current Status

Component	Status
Workload Integration	Ongoing
Pricing Model	Ongoing
Optimization Model	Ongoing
Algorithm Comparison	Ongoing
Baseline Comparison	Ongoing

8. Challenges Faced

1. Handling uncertainty in job duration prediction.
2. Synchronizing heterogeneous datasets (time alignment).
3. Modeling realistic SLA penalty structures.
4. Forecast accuracy limitations in carbon intensity signals.

9. Expected Outcomes

- A validated cost-aware carbon scheduling framework.
- Quantified trade-off curves between carbon savings and cost.
- Practical insights for sustainable cloud infrastructure management.
- Publication-ready experimental validation using real-world traces.